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use of only rigid circuits. Further development of the industry is expected to occur through the development of suitable processes that facilitate the cost-efficient manufacture of flexible electronics components and the integration of rigid components customised for various applications.

Budding electrical application areas, including heaters, solar cells, batteries and lighting, are propelling the growth of the worldwide flexible electronics market. The technologies enable electronic systems to be rolled, stretched, washed and worn, and thereby conform to various architectural features. Flexible electronics components are also capable of biodegradation with multiple-layer structures, which reduce the area required and number of interconnections, while increasing reliability of the system.

Key areas of application for flexible electronics include use in portable devices to provide a compact and robust form factor for electronic devices as well as medical applications like the creation of a circuit that exactly fits the gyri of the brain and enables the real-time study of epileptic patients. Other applications include the incorporation of flexible circuits-based lightweight impact sensors into the uniforms and helmets of soldiers to allow medics in providing better treatment to wounded soldiers. Stretchable circuits are also capable of being utilised with a standard balloon catheter to provide a range of radio circuitry and sensors to monitor electrical activity, pressure, temperature as well as blood flow inside the patient's body.

Organic electronic materials' market is poised to display exponential growth driven by increasing applications in conventional and niche areas, along with expanding in numerous applications ranging from wide-area displays and RFID tags to memories. Low cost for fabrication and high degree of flexibility are the two key factors driving the demand for organic electronics and its increasing penetration in the mainstream market. The materials are also useful for new and innovative applications, such as electronic paper and smart windows, wherein inorganic conductors (silicon or copper) cannot be employed. Organic conductors are also expected to find application in emerging frontiers, such as molecular computing.

Printed electronics, a new industry in its embryonic stage today, can shape the future of printing, as well as just about every other industry on the planet. Printed electronics is the application of conductive and electronic components onto a material by means of a printing process. It might not occur on a conventional press, but the applied product will be liquid or semi-liquid in form, and will cover a substrate pattern. ●

Concluded

The author is a final-year student of KMIT affiliated to JNTU/H. This article is based on a paper he prepared for a technical seminar on the subject